

Intro to NLP

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What is NLP

Definition:

- NLP stands for Natural Language Processing. It is a field of artificial intelligence (AI) that focuses on enabling computers to understand, interpret, generate, and interact using human language.

Importance:

- Chatbots and virtual assistants (e.g., Siri, Alexa)
- Language translation (e.g., Google Translate)
- Email spam filtering
- Search engines
- Voice recognition

problem in machine learning

- representation of text data that not have few unique texts

Content

- **Intro**
- **Classical NLP**
 - **Tokenization**
 - **Lemmatization**
 - **Stemming**
 - **Stop words**
 - **Bag of words**
 - **TD IDF**
 - **unigram**
 - **Bi gram**
 - **N gram**
 - **Gensim**
 - **Word2Vec**
 - **Avg Word2Vec**



Content

- **Deep learning**
 - **ANN**
 - **RNN**
 - **LSTM**
 - **GRU**
 - **BI directional LSTM RNN**
- **MAIN NLP**
 - **word embedding**
 - **word2vec**
 - **Encoders & Decoders**
 - **Attention Models**



Content

- **advanced NLP**
 - **Transformers**
 - **Bert**
 - **LLM**
 - **Prompt Engineering**
 - **Hugging Face**
 - **Fine Tuning**
- **Advanced Topics**
 - **RAG**
 - **Graph RAG**
 - **Langchain**
 - **Langgraph (state graph management)**
 - **Agent with tools (thought act)**

